GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2024.

GATE 2024

GATE 2024 | Q.1 | 1 mark

If '$\rightarrow$' denotes increasing order of intensity, then the meaning of the words [charm $\rightarrow$ enamor $\rightarrow$ bewitch] is analogous to [bored $\rightarrow$ _____ $\rightarrow$ weary]. Which one of the given options is appropriate to fill the blank?

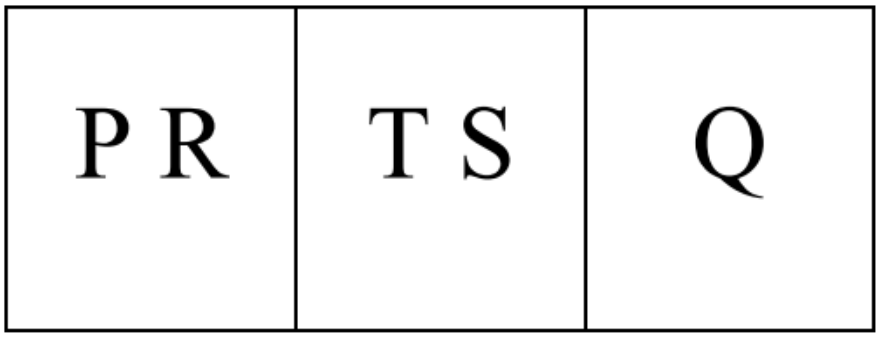
(A) jaded
(B) baffled
(C) dead
(D) worsted

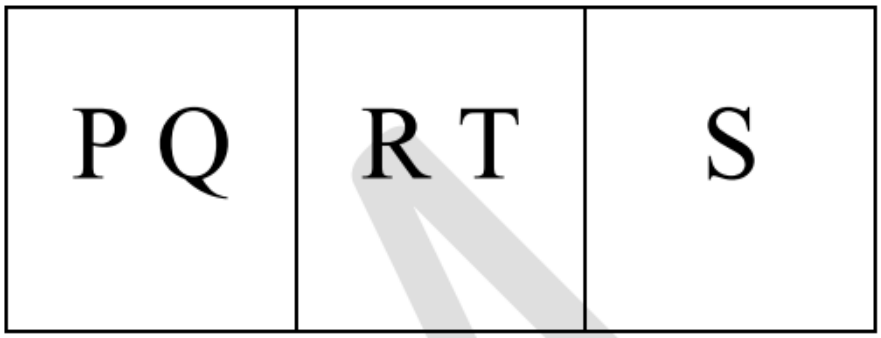
Answer: (A) [self-solved, no official key available]

Solution. Step 1. The first group lists words of increasing intensity: charm, enamor, bewitch.
Step 2. For bored $\rightarrow$? $\rightarrow$ weary, the middle word must be stronger than *bored* and weaker than *weary*: *jaded*.

GATE 2024 | Q.2 | 1 mark

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room. If S and Q are single children, and the room allocations shown below are acceptable to all, then, which one of the given options is the siblings?





(A) P and T
(B) P and S
(C) T and Q
(D) T and R

Answer: (A) [self-solved, no official key available]

Solution. Step 1. S and Q are single children, so the siblings are two of P, R, T.
Step 2. The siblings never share a room in either acceptable allocation shown. Checking the two allocations leaves only the pair P, T as always separated.

GATE 2024 | Q.3 | 1 mark

Five years ago, the ratio of Aman's age to his father's age was 1 : 4, and five years from now, the ratio will be 2 : 5. What was his father's age when Aman was born?

(A) 28 years
(B) 30 years
(C) 35 years
(D) 32 years

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Let Aman be a and the father f years old now: $\frac{a-5}{f-5} = \frac{1}{4}$ and $\frac{a+5}{f+5} = \frac{2}{5}$.
Step 2. $f = 4a - 15$ and $5a + 25 = 2f + 10 = 8a - 20 \Rightarrow a = 15$, $f = 45$.
Step 3. Father's age at Aman's birth $= 45 - 15 = 30$ years.

GATE 2024 | Q.4 | 1 mark

For a real number $x > 1$, $\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$. The value of x is

(A) 4
(B) 12
(C) 24
(D) 36

Answer: (C) [self-solved, no official key available]

Solution. Step 1. $\frac{1}{\log_b x} = \log_x b$.
Step 2. $\log_x 2 + \log_x 3 + \log_x 4 = \log_x 24 = 1$.
Step 3. $x = 24$.

GATE 2024 | Q.5 | 1 mark

The greatest prime factor of $\left(3^{199} - 3^{196}\right)$ is

(A) 13
(B) 17
(C) 3
(D) 11

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $3^{199} - 3^{196} = 3^{196}(27 - 1) = 3^{196} \times 26$.
Step 2. $26 = 2 \times 13$.
Step 3. The greatest prime factor is 13.

GATE 2024 | Q.6 | 2 marks

Sequence the following sentences (P, Q, R, S) in a coherent passage:
P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?"
Q: Shifu said, "Surely my running away from the bull is also an illusion."
R: Shifu once proclaimed that all life is illusion.
S: One day, when a bull gave him chase, Shifu began running for his life.

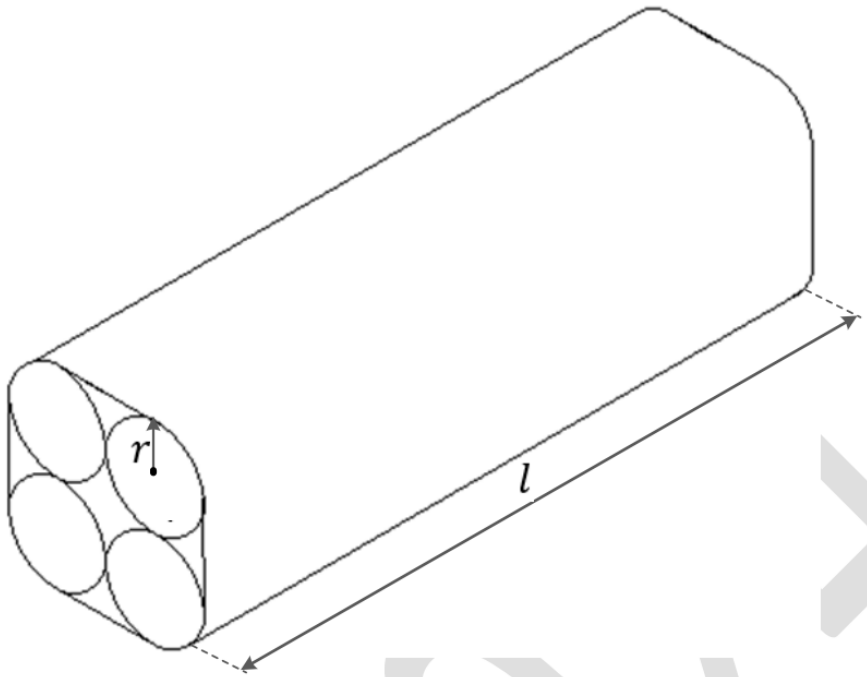
(A) SPRQ
(B) SRPQ
(C) RSPQ
(D) RPQS

Answer: (C) [self-solved, no official key available]

Solution. Step 1. R states the premise (life is illusion), so it comes first.
Step 2. S gives the incident (the bull), P the student's question and Q Shifu's reply: $R\,S\,P\,Q$.

GATE 2024 | Q.7 | 2 marks

Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown in the following figure. The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once, is



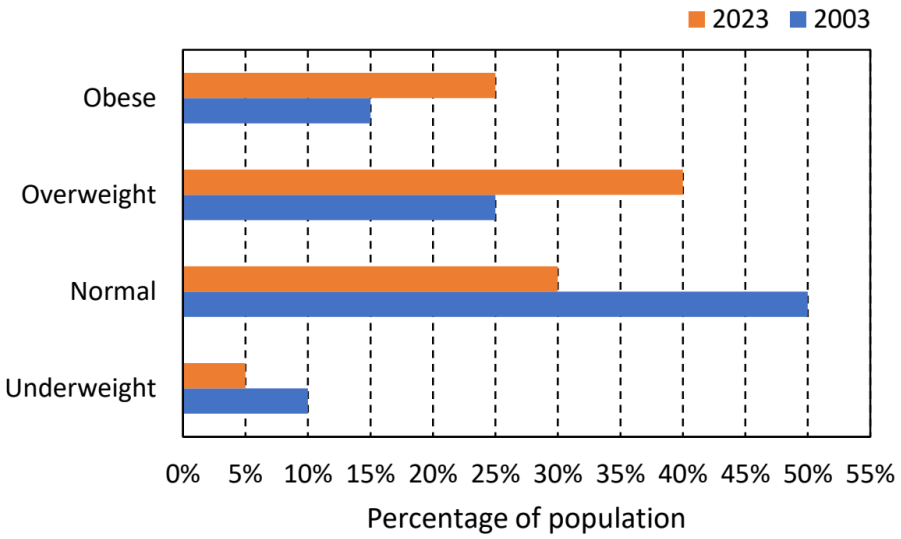
(A) $20(4+\pi)$
(B) $20(8+\pi)$
(C) $10(8+\pi)$
(D) $10(4+\pi)$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. The four cylinders touch in a square arrangement; the belt runs along four straight pieces of length $2r = 1$ cm and four quarter-circular arcs of radius r.
Step 2. Perimeter $= 4(2r) + 2\pi r = 4 + \pi$ cm.
Step 3. Area $=$ perimeter $\times$ width $= (4+\pi) \times 10 = 10(4+\pi)$ cm^2.

GATE 2024 | Q.8 | 2 marks

The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023. Based on the data provided, which one of the following options is INCORRECT?



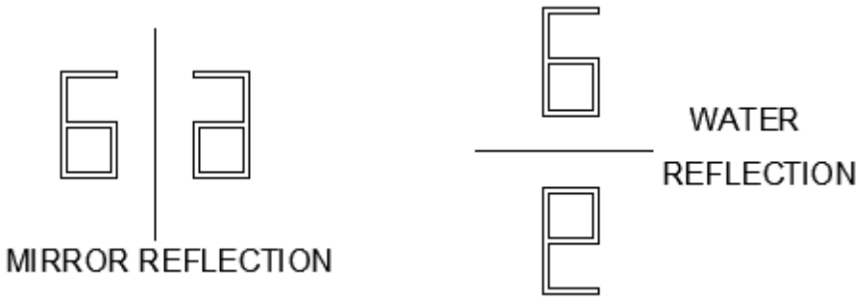
(A) The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.
(B) The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.
(C) The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.
(D) The percentage of population falling into normal category has decreased in 20 years.

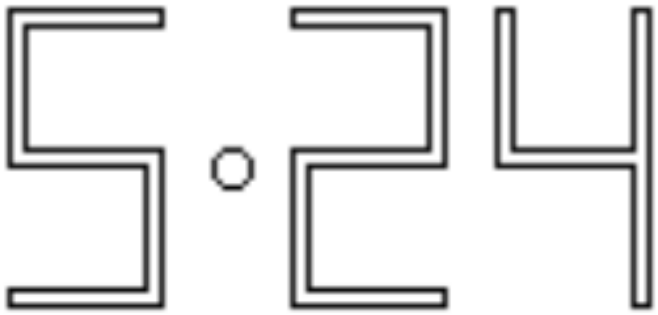
Answer: (C) [self-solved, no official key available]

Solution. Step 1. Read off the chart: 2003 – underweight 10, normal 50, overweight 25, obese 15; 2023 – underweight 5, normal 30, overweight 40, obese 25 (%).
Step 2. Obese/normal: 0.3 (2003) $\rightarrow$ 0.83 (2023): it has *increased*, so statement (C) is incorrect.
Step 3. (A) 0.5 $\rightarrow$ 1.33 (true); (B) 0.2 $\rightarrow$ 0.17 (true); (D) 50 $\rightarrow$ 30 (true).

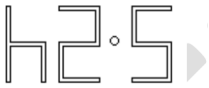
GATE 2024 | Q.9 | 2 marks

Examples of mirror and water reflections are shown in the figures below. An object appears as the following image after first reflecting in a mirror and then reflecting on water. The original object is





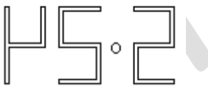
(A)



(B)



(C)



(D)



Answer: (A) [self-solved, no official key available]

Solution. Step 1. A mirror reflection flips the object left ↔ right; a water reflection flips it top ↔ bottom.
Step 2. The two together equal a rotation by 180°. Undoing it means rotating the final image by 180°, which gives option (A).

GATE 2024 | Q.10 | 2 marks

Two identical sheets A and B, of dimensions 24 cm × 16 cm, can be folded into half using two distinct operations, FO1 or FO2. In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge. If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

(A) 14:11
(B) 11:14
(C) 18:11
(D) 11:18

Answer: (A) [self-solved, no official key available]

Solution. Step 1. FO1 halves the short side (16) each time: after two folds A is 24×4, perimeter $= 2(24 + 4) = 56$.
Step 2. FO2 halves the long side (24) each time: after two folds B is 6×16, perimeter $= 2(6 + 16) = 44$.
Step 3. Ratio $56 : 44 = 14 : 11$.

GATE 2024 | Q.11 | 1 mark

The general form of the complementary function of a differential equation is given by $y(t) = (At + B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is _____.

(A) $\dfrac{d^2y}{dt^2} + 4\dfrac{dy}{dt} + 4y = f(t)$

(B) $\dfrac{d^2y}{dt^2} + 4y = f(t)$

(C) $\dfrac{d^2y}{dt^2} + 3\dfrac{dy}{dt} + 2y = f(t)$

(D) $\dfrac{d^2y}{dt^2} + 5\dfrac{dy}{dt} + 6y = f(t)$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $(At + B)e^{-2t}$ corresponds to a repeated characteristic root $\lambda = -2$.
Step 2. Characteristic equation: $(\lambda + 2)^2 = \lambda^2 + 4\lambda + 4 = 0$.
Step 3. The differential equation is $y'' + 4y' + 4y = f(t)$.

GATE 2024 | Q.12 | 1 mark

In the context of Bode magnitude plots, 40 dB/decade is the same as _____.

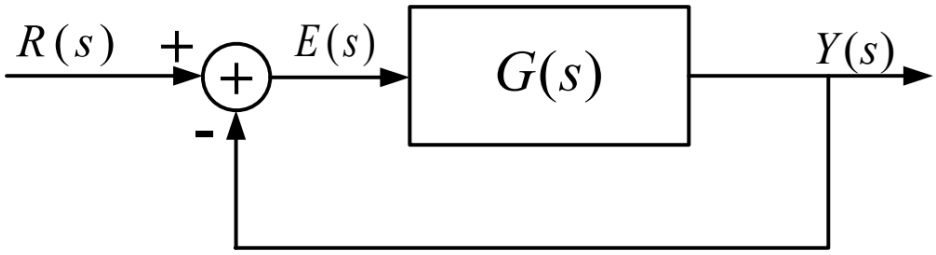
(A) 12 dB/octave
(B) 6 dB/octave
(C) 20 dB/octave
(D) 10 dB/octave

Answer: (A) [self-solved, no official key available]

Solution. Step 1. One octave is a factor 2 in frequency $= \log_{10} 2 = 0.301$ decade.
Step 2. 40 dB/dec $\times$ 0.301 = 12 dB/octave.

GATE 2024 | Q.13 | 1 mark

In the feedback control system shown in the figure below, $G(s) = \dfrac{6}{s(s+1)(s+2)}$. $R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, respectively. If the input $r(t)$ is a unit step function, then _____.



(A) $\lim_{t\to\infty} e(t) = 0$

(B) $\lim_{t\to\infty} e(t) = \dfrac{1}{3}$

(C) $\lim_{t\to\infty} e(t) = \dfrac{1}{4}$

(D) $\lim_{t\to\infty} e(t)$ does not exist, $e(t)$ is oscillatory

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Closed-loop characteristic equation: $s(s+1)(s+2) + 6 = s^3 + 3s^2 + 2s + 6 = 0$.
Step 2. Routh array: row s^1 has $\dfrac{3 \cdot 2 - 6}{3} = 0$ and the row s^2: $3s^2 + 6 = 0$ gives roots $s = \pm j\sqrt{2}$.
Step 3. There are poles on the imaginary axis, so $e(t)$ is oscillatory and the limit does not exist.

GATE 2024 | Q.14 | 1 mark

A digital communication system transmits through a noiseless bandlimited channel $[-W, W]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum_n b[n]\, x(t - nT)$, where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = mT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

(A) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T$

(B) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = 1$

(C) $\sum_{m=-\infty}^{\infty} X(f + mT) = T$

(D) $\sum_{m=-\infty}^{\infty} X(f + mT) = 1$

Answer: (A) [self-solved, no official key available]

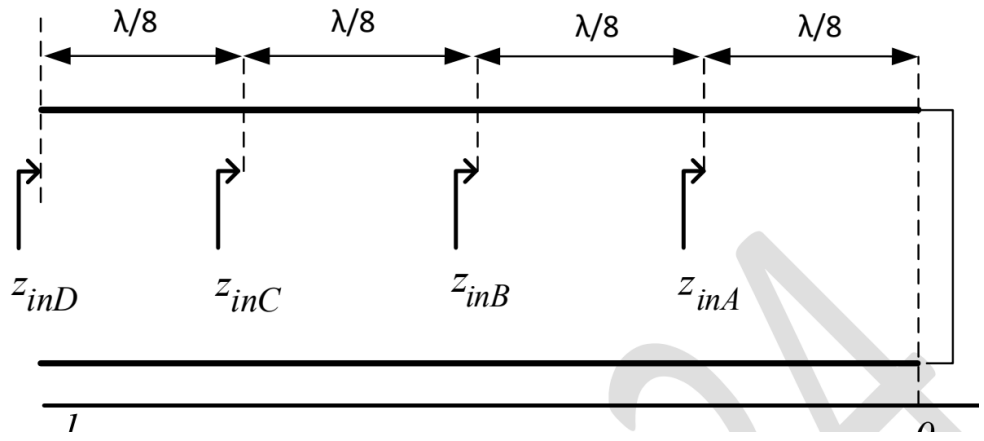
Solution. Step 1. Zero ISI requires $x(mT) = \delta[m]$.

Step 2. Sampling $x(t)$ at rate $1/T$ gives the periodic spectrum $\frac{1}{T}\sum_m X\left(f - \frac{m}{T}\right)$, which must equal the transform of $\delta[m]$, a constant 1.

Step 3. Hence $\sum_m X\left(f + \frac{m}{T}\right) = T$.

GATE 2024 | Q.15 | 1 mark

Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances z_{inA}, z_{inB}, z_{inC}, and z_{inD} (indicated in the figure) are _____.



(A) $z_{inA} = +1j,\ z_{inB} = \infty,\ z_{inC} = -1j,\ z_{inD} = 0$

(B) $z_{inA} = \infty,\ z_{inB} = +0.4j,\ z_{inC} = 0,\ z_{inD} = +0.4j$

(C) $z_{inA} = -1j,\ z_{inB} = 0,\ z_{inC} = +1j,\ z_{inD} = \infty$

(D) $z_{inA} = +0.4j,\ z_{inB} = \infty,\ z_{inC} = -0.4j,\ z_{inD} = 0$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. For a short-circuited line, $z_{in} = j \tan \beta d$ with $\beta d = \frac{2\pi}{\lambda} d$.

Step 2. At $d = \lambda/8$: $j \tan 45^\circ = +j$; at $\lambda/4$: ∞; at $3\lambda/8$: $j \tan 135^\circ = -j$; at $\lambda/2$: 0.

GATE 2024 | Q.16 | 1 mark

Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

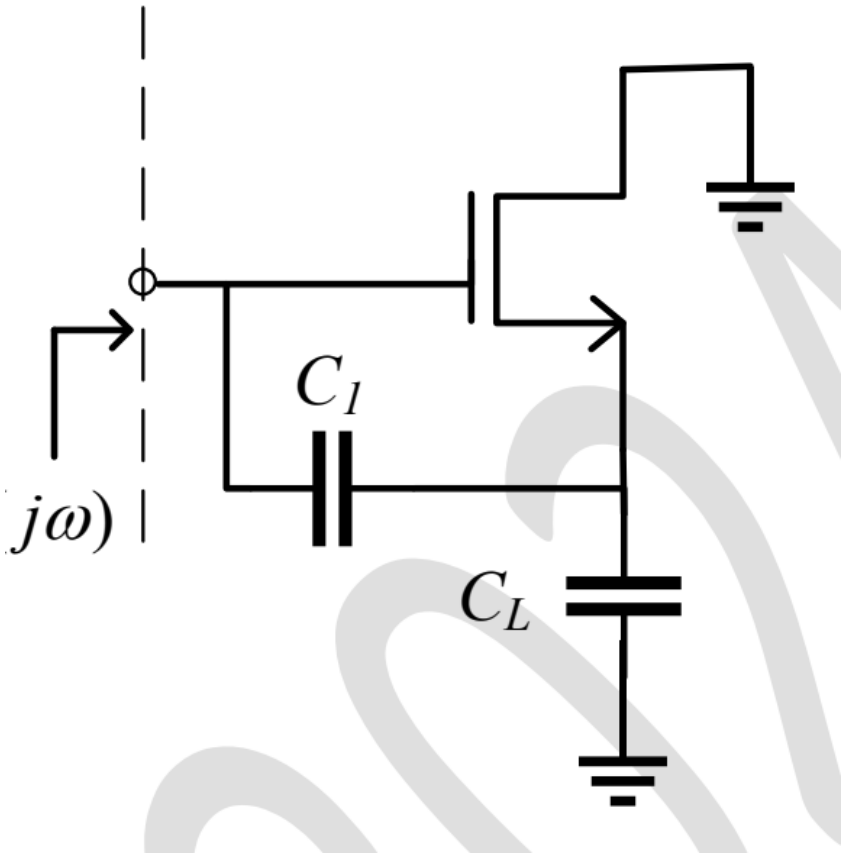
(A) Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.
(B) Only $\vec{F}_1$ is an electrostatic field.
(C) Only $\vec{F}_2$ is an electrostatic field.
(D) Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

Answer: (B) [self-solved, no official key available]

Solution. Step 1. An electrostatic field is irrotational: $\nabla \times \vec{F} = 0$.
Step 2. $\vec{F}_1$: $(\nabla \times \vec{F}_1)_z = \dfrac{\partial(Ax)}{\partial x} - \dfrac{\partial(Ay)}{\partial y} = A - A = 0$.
Step 3. $\vec{F}_2$: $(\nabla \times \vec{F}_2)_z = -A - A = -2A \neq 0$.
Step 4. Only $\vec{F}_1$ is electrostatic.

GATE 2024 | Q.17 | 1 mark

In the circuit below, assume that the long channel NMOS transistor is biased in saturation. The small signal trans-conductance of the transistor is g_m. Neglect body effect, channel length modulation and intrinsic device capacitances. The small signal input impedance $Z_{in}(j\omega)$ is _____.



(A) $\dfrac{-g_m}{\omega^2 C_1 C_L} + \dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$
(B) $\dfrac{g_m}{\omega^2 C_1 C_L} + \dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$
(C) $\dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$
(D) $\dfrac{-g_m}{\omega^2 C_1 C_L} + \dfrac{1}{j\omega C_1 + j\omega C_L}$

Answer: (A) [self-solved, no official key available]

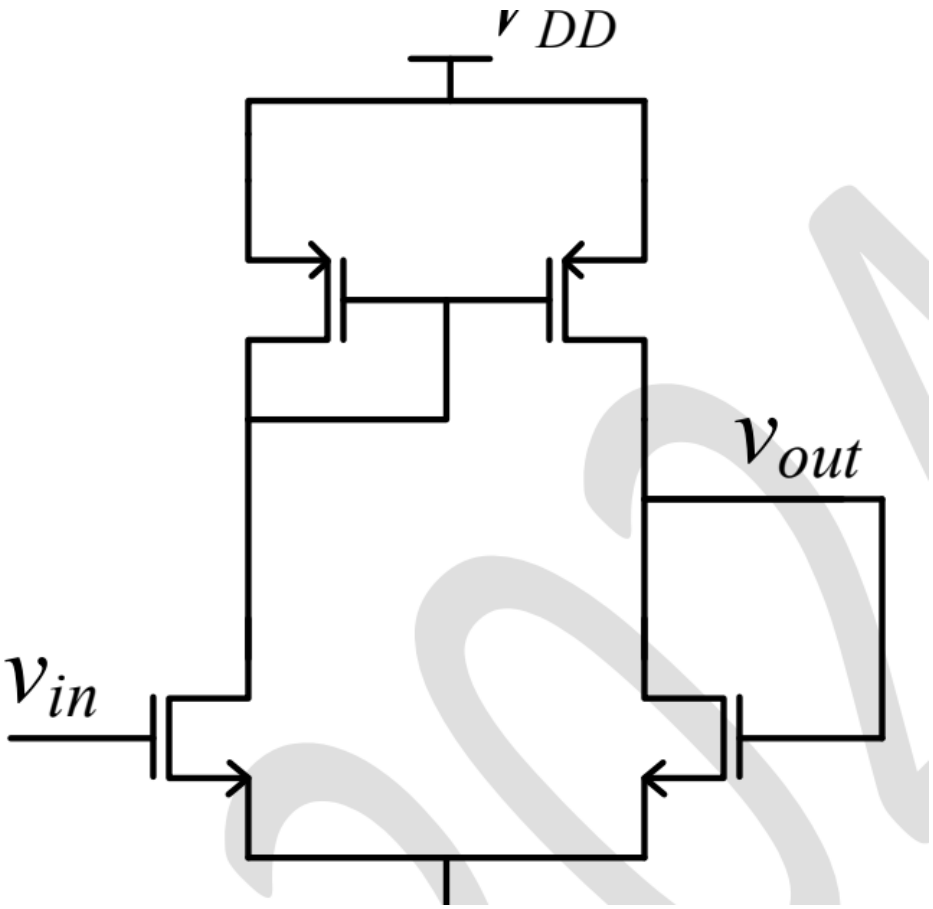
Solution. Step 1. The input current i flows through C_1 into the gate/source network. The voltage gate-to-source is $v_{gs} = i/(j\omega C_1)$ and the transistor current $g_m v_{gs}$ also flows through C_L.

Step 2. Summing the voltages: $v_{in} = \dfrac{i}{j\omega C_1} + \dfrac{i}{j\omega C_L} + \dfrac{g_m i}{j\omega C_1 \cdot j\omega C_L}$.

Step 3. Since $(j\omega)^2 = -\omega^2$, $Z_{in} = \dfrac{-g_m}{\omega^2 C_1 C_L} + \dfrac{1}{j\omega C_1} + \dfrac{1}{j\omega C_L}$: a negative resistance in series with the capacitors.

GATE 2024 | Q.18 | 1 mark

For the closed loop amplifier circuit shown below, the magnitude of open loop low frequency small signal voltage gain is 40. All the transistors are biased in saturation. The current source I_{SS} is ideal. Neglect body effect, channel length modulation and intrinsic device capacitances. The closed loop low frequency small signal voltage gain $\dfrac{v_{out}}{v_{in}}$ (rounded off to three decimal places) is _____.



(A) 0.976
(B) 1.000
(C) 1.025
(D) 0.488

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The output is fed back directly to the inverting input (unity feedback, $\beta = 1$).

Step 2. $A_{cl} = \dfrac{A}{1 + A\beta} = \dfrac{40}{41} = 0.976$.

GATE 2024 | Q.19 | 1 mark

For the Boolean function $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$, the essential prime implicants are _____.

(A) BD, $\overline{B}\,\overline{D}$
(B) BD, AB
(C) AB, $\overline{B}\,\overline{D}$
(D) BD, $\overline{B}\,\overline{D}$, AB

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Prime implicants from the K-map: $\overline{B}\,\overline{D}$ (covers $0, 2, 8, 10$), BD (covers $5, 7, 13, 15$), AB (covers $12, 13, 14, 15$).
Step 2. Minterm 0 is covered only by $\overline{B}\,\overline{D}$; 7 only by BD; 12 and 14 only by AB.
Step 3. All three are essential.

GATE 2024 | Q.20 | 1 mark

A white Gaussian noise $w(t)$ with zero mean and power spectral density $\dfrac{N_0}{2}$, when applied to a first-order RC low pass filter produces an output $n(t)$. At a particular time $t = t_k$, the variance of the random variable $n(t_k)$ is _____.

(A) $\dfrac{N_0}{4RC}$
(B) $\dfrac{N_0}{2RC}$
(C) $\dfrac{N_0}{RC}$
(D) $\dfrac{N_0}{2}RC$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. $h(t) = \dfrac{1}{RC}e^{-t/RC}u(t)$.
Step 2. $\sigma^2 = \dfrac{N_0}{2}\int_0^\infty h^2(t)\,dt = \dfrac{N_0}{2}\cdot\dfrac{1}{(RC)^2}\cdot\dfrac{RC}{2} = \dfrac{N_0}{4RC}$.

GATE 2024 | Q.21 | 1 mark

A causal and stable LTI system with impulse response $h(t)$ produces an output $y(t)$ for an input signal $x(t)$. A signal $x(0.5t)$ is applied to another causal and stable LTI system with impulse response $h(0.5t)$. The resulting output is _____.

(A) $2y(0.5t)$
(B) $4y(0.5t)$
(C) $0.25\,y(2t)$
(D) $0.25\,y(0.25t)$

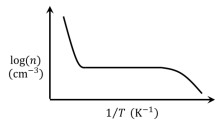
Answer: (A) [self-solved, no official key available]

Solution. Step 1. Output $= \int x(0.5\tau)\,h(0.5(t-\tau))\,d\tau$. Substitute $u = 0.5\tau$, $d\tau = 2\,du$.
Step 2. $= 2\int x(u)\,h(0.5t-u)\,du = 2\,y(0.5t)$.

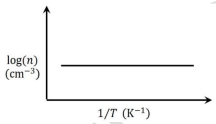
GATE 2024 | Q.22 | 1 mark

For non-degenerately doped n-type silicon, which one of the following plots represents the temperature (T) dependence of free electron concentration (n)?

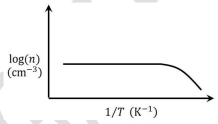
(A)



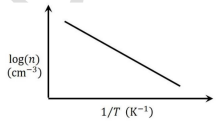
(B)



(C)



(D)

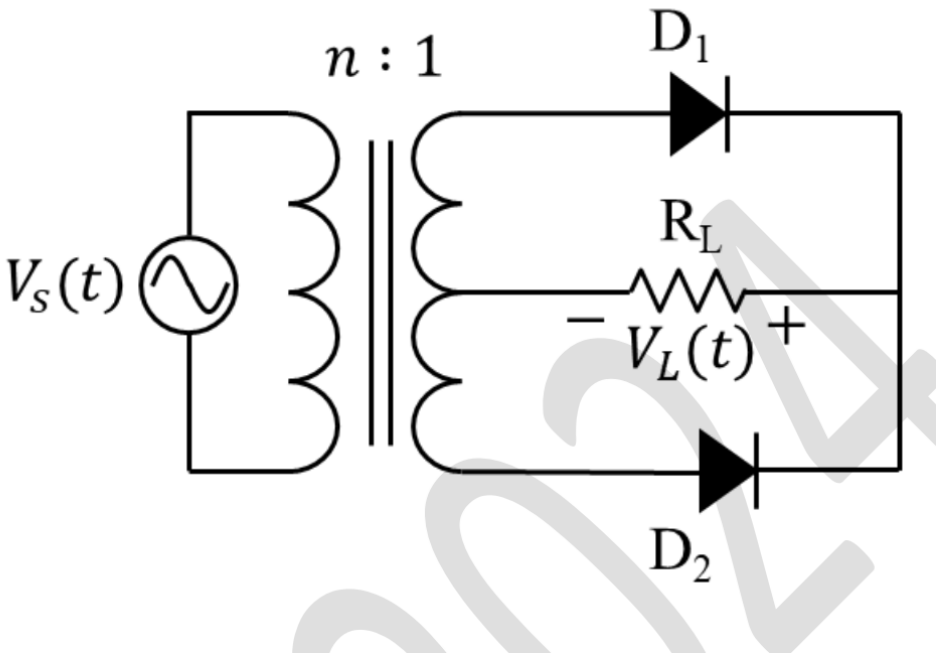


Answer: (A) [self-solved, no official key available]

Solution. Step 1. At low temperature (large $1/T$) carriers freeze out, so $\log n$ falls.
Step 2. In the extrinsic range $n \approx N_D$ (flat).
Step 3. At high temperature intrinsic carriers dominate and $\log n$ rises sharply (small $1/T$). The plot of $\log(n)$ vs $1/T$ showing this behaviour is (A).

GATE 2024 | Q.23 | 1 mark

In the circuit shown, the $n:1$ step-down transformer and the diodes are ideal. The diodes have no voltage drop in forward biased condition. If the input voltage (in Volts) is $V_s(t) = 10\sin\omega t$ and the average value of load voltage $V_L(t)$ (in Volts) is $\dfrac{2.5}{\pi}$, the value of n is _____.



(A) 4
(B) 8
(C) 12
(D) 16

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The secondary peak is $10/n$ and the rectifier is full-wave, so $V_{L,avg} = \dfrac{2}{\pi}\cdot\dfrac{10}{n}$.
Step 2. $\dfrac{20}{n\pi} = \dfrac{2.5}{\pi} \Rightarrow n = 8$.

GATE 2024 | Q.24 | 1 mark

For a causal discrete-time LTI system with transfer function $H(z) = \dfrac{2z^2+3}{\left(z+\frac{1}{3}\right)\left(z-\frac{1}{3}\right)}$ which of the following statements is/are true?

(A) The system is stable.
(B) The system is a minimum phase system.
(C) The initial value of the impulse response is 2.
(D) The final value of the impulse response is 0.

Answer: (A), (C), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. Poles at $\pm\frac{1}{3}$ are inside the unit circle and the system is causal $\Rightarrow$ stable (A) and $h[n] \to 0$ (D).
Step 2. Zeros: $2z^2+3=0 \Rightarrow z = \pm j\sqrt{1.5}$, $|z| = 1.22 > 1$: outside the unit circle $\Rightarrow$ not minimum phase (B false).
Step 3. $h[0] = \lim_{z\to\infty} H(z) = 2$ (C).

GATE 2024 | Q.25 | 1 mark

Let $\rho(x,y,z,t)$ and $\mathbf{u}(x,y,z,t)$ represent density and velocity, respectively, at a point (x,y,z) and time t. Assume $\dfrac{\partial\rho}{\partial t}$ is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\hat{n}$ be the outward unit normal of S. Which of the following equations is/are equivalent to $\dfrac{\partial\rho}{\partial t} + \nabla\cdot(\rho\mathbf{u}) = 0$?

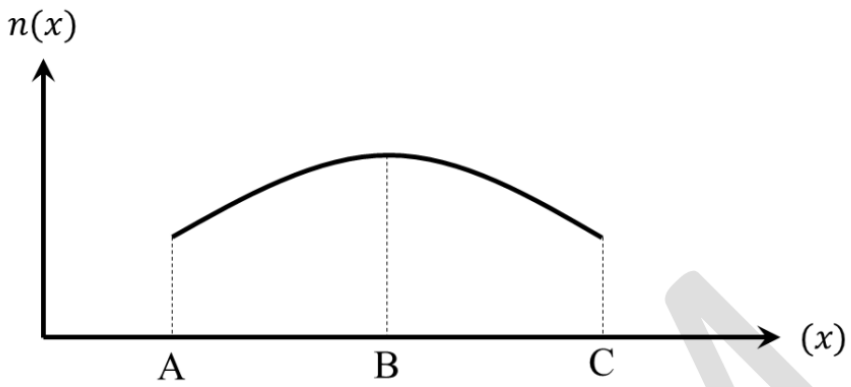
(A) $\dfrac{d}{dt}\displaystyle\int_V \rho\, dv = -\oint_S \rho\,\mathbf{u}\cdot\hat{n}\, ds$
(B) $\dfrac{d}{dt}\displaystyle\int_V \rho\, dv = \oint_S \rho\,\mathbf{u}\cdot\hat{n}\, ds$
(C) $\displaystyle\int_V \frac{\partial\rho}{\partial t} dv = -\int_V \nabla\cdot(\rho\mathbf{u})\, dv$
(D) $\displaystyle\int_V \frac{\partial\rho}{\partial t} dv = \int_V \nabla\cdot(\rho\mathbf{u})\, dv$

Answer: (A), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. Integrate the continuity equation over an arbitrary volume: $\int_V \frac{\partial\rho}{\partial t} dv = -\int_V \nabla\cdot(\rho\mathbf{u})\, dv$ (C).
Step 2. By the divergence theorem the right side is $-\oint_S \rho\mathbf{u}\cdot\hat{n}\, ds$ and the left side equals $\frac{d}{dt}\int_V \rho\, dv$ (A).

GATE 2024 | Q.26 | 1 mark

The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?



(A) For x between B and C, the electron diffusion current is directed from C to B.
(B) For x between B and A, the electron drift current is directed from B to A.
(C) For x between B and C, the electric field is directed from B to C.
(D) For x between B and A, the electric field is directed from A to B.

Answer: (A), (B), (C) (multiple select) [self-solved, no official key available]

Solution. Step 1. Between B and C the concentration falls with x: electrons diffuse from B to C, so the diffusion (conventional) current is directed from C to B (A true).
Step 2. At equilibrium the net current is zero, so the drift current is opposite: from B to C; electrons then drift against the field, so the field points from B to C (C true).
Step 3. Between B and A the concentration rises with x: diffusion current from A to B, drift current from B to A (B true), field from B to A (D false).

GATE 2024 | Q.27 | 1 mark

A machine has a 32-bit architecture with 1-word long instructions. It has 24 registers and supports an instruction set of size 40. Each instruction has five distinct fields, namely opcode, two source register identifiers, one destination register identifier, and an immediate value. Assuming that the immediate operand is an unsigned integer, its maximum value is _____.

Answer: 2047 [self-solved, no official key available]

Solution. Step 1. Opcode: $\lceil \log_2 40 \rceil = 6$ bits.
Step 2. Each register identifier: $\lceil \log_2 24 \rceil = 5$ bits, three of them: 15 bits.
Step 3. Immediate field: $32 - 6 - 15 = 11$ bits, maximum unsigned value $2^{11} - 1 = 2047$.

GATE 2024 | Q.28 | 1 mark

An amplitude modulator has output (in Volts) $s(t) = A\cos(400\pi t) + B\cos(360\pi t) + B\cos(440\pi t)$. The carrier power normalized to 1 Ω resistance is 50 Watts. The ratio of the total sideband power to the total power is $\frac{1}{9}$. The value of B (in Volts, rounded off to two decimal places) is _____.

Answer: 2.50 [self-solved, no official key available]

Solution. Step 1. Carrier power $\frac{A^2}{2} = 50 \Rightarrow A = 10$ V.
Step 2. Total sideband power $= 2 \cdot \frac{B^2}{2} = B^2$.
Step 3. $\frac{B^2}{50 + B^2} = \frac{1}{9} \Rightarrow 9B^2 = 50 + B^2 \Rightarrow B^2 = 6.25$, $B = 2.50$ V.

GATE 2024 | Q.29 | 1 mark

In a number system of base r, the equation $x^2 - 12x + 37 = 0$ has $x = 8$ as one of its solutions. The value of r is _____.

Answer: 11 [self-solved, no official key available]

Solution. Step 1. In base r: $12_r = r + 2$ and $37_r = 3r + 7$.
Step 2. $8^2 - (r + 2) \cdot 8 + (3r + 7) = 0 \Rightarrow 64 - 8r - 16 + 3r + 7 = 0$.
Step 3. $55 = 5r \Rightarrow r = 11$.

GATE 2024 | Q.30 | 1 mark

Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors $\{[2 \quad -3 \quad \alpha],\ [3 \quad -1 \quad 3],\ [1 \quad -5 \quad 7]\}$ does not form a basis of $\mathbb{R}^3$ is

_____.

Answer: 5 [self-solved, no official key available]

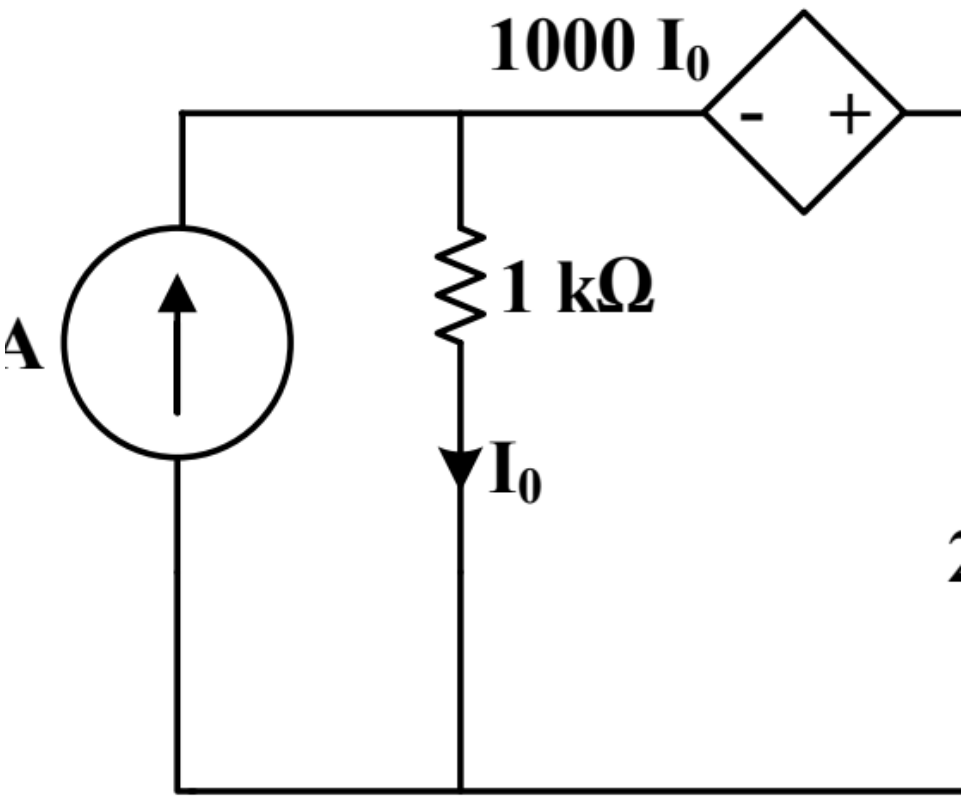
Solution. Step 1. The set is not a basis iff the three vectors are linearly dependent: $\det = 0$.

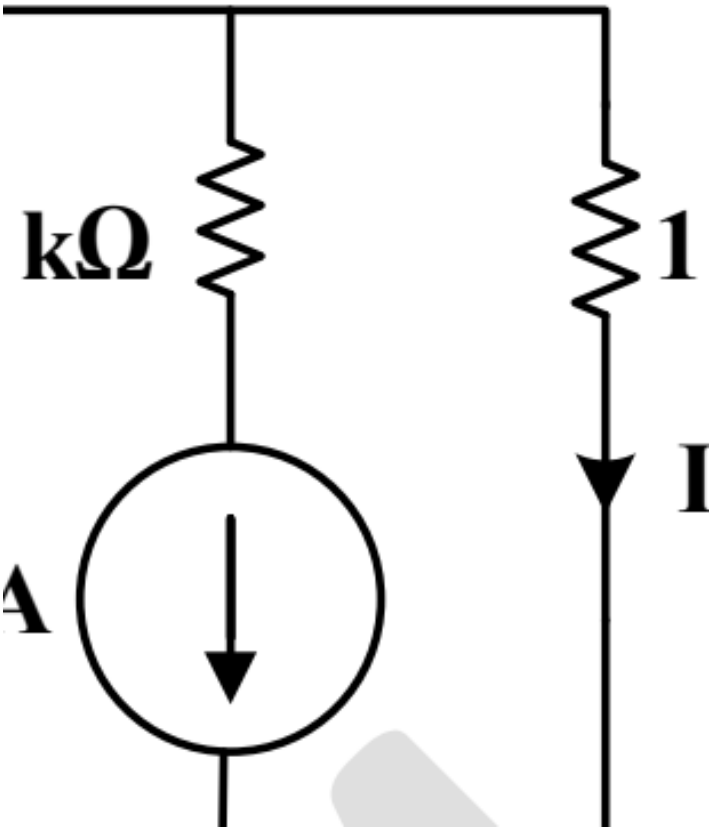
Step 2. $\begin{vmatrix} 2 & -3 & \alpha \\ 3 & -1 & 3 \\ 1 & -5 & 7 \end{vmatrix} = 2(-7+15) + 3(21-3) + \alpha(-15+1) = 16 + 54 - 14\alpha.$

Step 3. $70 - 14\alpha = 0 \Rightarrow \alpha = 5.$

GATE 2024 | Q.31 | 1 mark

In the given circuit, the current I_x (in mA) is _____.





Answer: 1.2 [self-solved, no official key available]

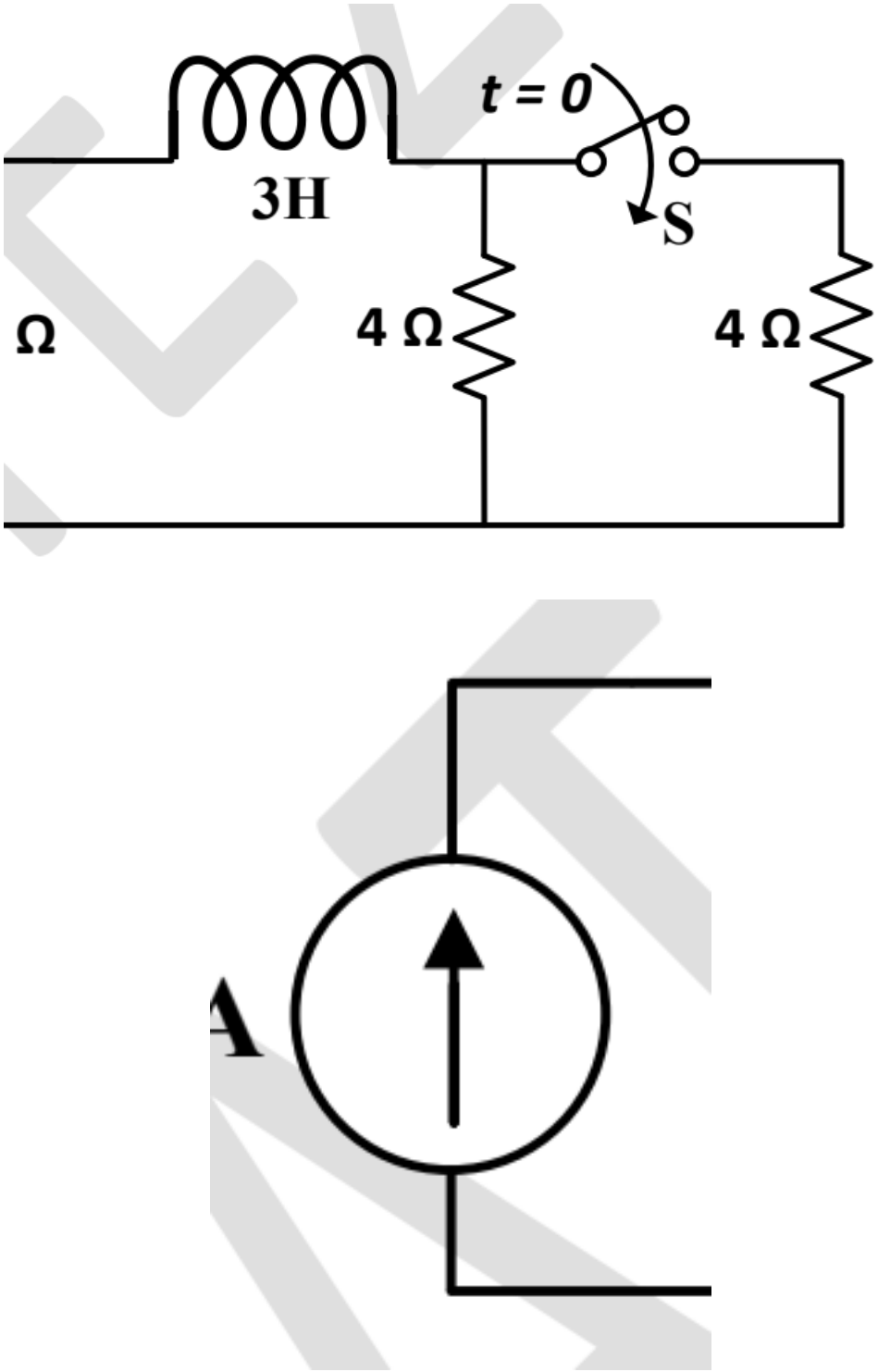
Solution. Step 1. Let I_0 be the current in the first 1 kΩ resistor; its voltage is $V_A = 1000I_0$. The dependent voltage source is $1000I_0$ in series, so the next node is $V_B = V_A + 1000I_0 = 2V_A$.

Step 2. KCL at the source node: $5 = I_0 + I_{AB}$ with $I_{AB} = \dfrac{V_B}{1} + 2 + \dfrac{V_B}{1}$ in mA, V, kΩ units, i.e. $5 - I_0 = 4I_0 + 2$.

Step 3. $I_0 = 0.6$ mA, $V_B = 2V_A = 1.2$ V, and $I_x = \dfrac{1.2}{1\ \text{k}\Omega} = 1.2$ mA.

GATE 2024 | Q.32 | 1 mark

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____.



Answer: 0.75 [self-solved, no official key available]

Solution. Step 1. After closing, find the resistance seen by the 3 H inductor with the current source open-circuited.
Step 2. The 2 Ω resistor in series with the parallel pair of 4 Ω resistors (2 Ω) gives $R_{eq} = 4\ \Omega$.
Step 3. $\tau = \dfrac{L}{R_{eq}} = \dfrac{3}{4} = 0.75$ s.

GATE 2024 | Q.33 | 1 mark

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval $[0, 1]$. The probability that $X \geq Y$ is _____.

Answer: 0.5 [self-solved, no official key available]

Solution. Step 1. X and Y are i.i.d. continuous, so $P(X > Y) = P(Y > X)$ and $P(X = Y) = 0$.
Step 2. Therefore $P(X \geq Y) = \frac{1}{2}$.

GATE 2024 | Q.34 | 1 mark

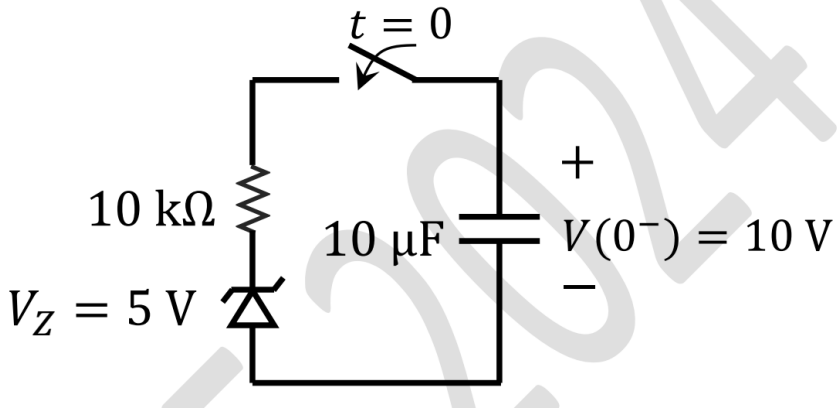
A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____.

Answer: 4 [self-solved, no official key available]

Solution. Step 1. Entropy is maximum for equiprobable symbols: $H_{max} = \log_2 M$.
Step 2. $\log_2 16 = 4$ bits.

GATE 2024 | Q.35 | 1 mark

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode ($V_Z = 5$ V) after the switch is closed (in mJ, rounded off to three decimal places) is _____.



Answer: 0.250 [self-solved, no official key available]

Solution. Step 1. The capacitor discharges through 10 kΩ and the Zener while the Zener conducts in breakdown, clamping its voltage at 5 V. Conduction stops when the capacitor voltage reaches 5 V.
Step 2. Charge passed through the Zener: $\Delta Q = C\,\Delta V = 10\ \mu\text{F} \times (10 - 5) = 50\ \mu\text{C}$.
Step 3. Energy in the Zener $= V_Z \Delta Q = 5 \times 50\ \mu\text{C} = 0.250$ mJ.

GATE 2024 | Q.36 | 2 marks

Consider the Earth to be a perfect sphere of radius R. Then the surface area of the region, enclosed by the 60°N latitude circle, that contains the north pole in its interior is _____.

(A) $\pi R^2(2 - \sqrt{3})$
(B) $\dfrac{\pi R^2(\sqrt{2} - 1)}{2}$
(C) $2\sqrt{3}\,\pi R^2$
(D) $\dfrac{8\pi R^2(2 - \sqrt{3})}{2}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. The polar cap down to colatitude $\theta_0 = 30°$ has area $2\pi R^2(1 - \cos\theta_0)$.
Step 2. $= 2\pi R^2 \left(1 - \dfrac{\sqrt{3}}{2}\right) = \pi R^2(2 - \sqrt{3})$.

GATE 2024 | Q.37 | 2 marks

Consider a unity negative feedback control system with forward path gain $G(s) = \dfrac{K}{(s+1)(s+2)(s+3)}$ as shown. The impulse response of the closed-loop system decays faster than e^{-t} if _____.

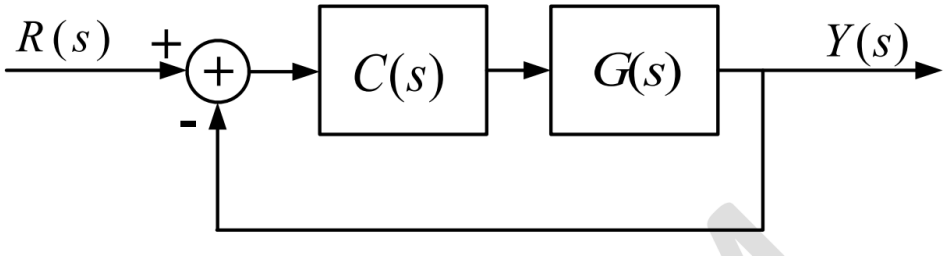
(A) $1 \le K \le 5$
(B) $7 \le K \le 21$
(C) $-4 \le K \le -1$
(D) $-24 \le K \le -6$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Closed-loop characteristic polynomial: $s^3 + 6s^2 + 11s + 6 + K$.
Step 2. Decay faster than e^{-t} means all poles have Re $s < -1$. Shift $s = p - 1$: $p^3 + 3p^2 + 2p + K$.
Step 3. Routh: all coefficients positive and $3 \cdot 2 > K$, so $0 < K < 6$; among the options $1 \le K \le 5$ lies inside this range.

GATE 2024 | Q.38 | 2 marks

A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = \dfrac{1}{s^2}$ cascaded with a compensator $C(s) = \dfrac{K(s+\alpha)}{s+4}$, where K and α are positive real constants. In order for the closed-loop system to have poles at $-1 \pm j\sqrt{3}$, the value of α must be _____.



(A) 0
(B) 1
(C) 2
(D) 3

Answer: (B) [self-solved, no official key available]

Solution. Step 1. Characteristic equation: $s^2(s+4) + K(s+\alpha) = s^3 + 4s^2 + Ks + K\alpha = 0$.
Step 2. Poles at $-1 \pm j\sqrt{3}$ give the factor s^2+2s+4; the third root $-r$: $(s^2+2s+4)(s+r) = s^3+(2+r)s^2+(4+2r)s+4r$.
Step 3. $2 + r = 4 \Rightarrow r = 2$; then $K = 4 + 2r = 8$ and $K\alpha = 4r = 8$, so $\alpha = 1$.

GATE 2024 | Q.39 | 2 marks

A uniform plane wave with electric field $\vec{E}(x) = A_y e^{-j3x}\hat{a}_y$ V/m is travelling in the air (relative permittivity, $\varepsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____.

(A) $-\dfrac{3\pi}{4}$
(B) $-\dfrac{3\pi}{2}$
(C) $-\dfrac{\pi}{6}$
(D) $-\dfrac{\pi}{3}$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The phase constant is $\beta = 3$ rad/m. At a perfect conductor the tangential E vanishes and the tangential H is doubled (maximum) at $x = 0$: $H \propto \cos \beta x$.

Step 2. The first null of $\cos(3x)$ occurs at $3x = -\frac{\pi}{2}$, i.e. $x = -\frac{\pi}{6}$ m.

GATE 2024 | Q.40 | 2 marks

A 4-bit priority encoder has inputs D_3, D_2, D_1, and D_0 in descending order of priority. The two-bit output AB is generated as $00, 01, 10$, and 11 corresponding to inputs D_3, D_2, D_1, and D_0, respectively. The Boolean expression of the output bit B is _____.

(A) $\overline{D_3}\,\overline{D_2}$
(B) $\overline{D_3}D_2 + \overline{D_3}\,\overline{D_1}$
(C) $D_3\overline{D_2} + \overline{D_3}D_1$
(D) $\overline{D_3}\,\overline{D_1}$

Answer: (B) [self-solved, no official key available]

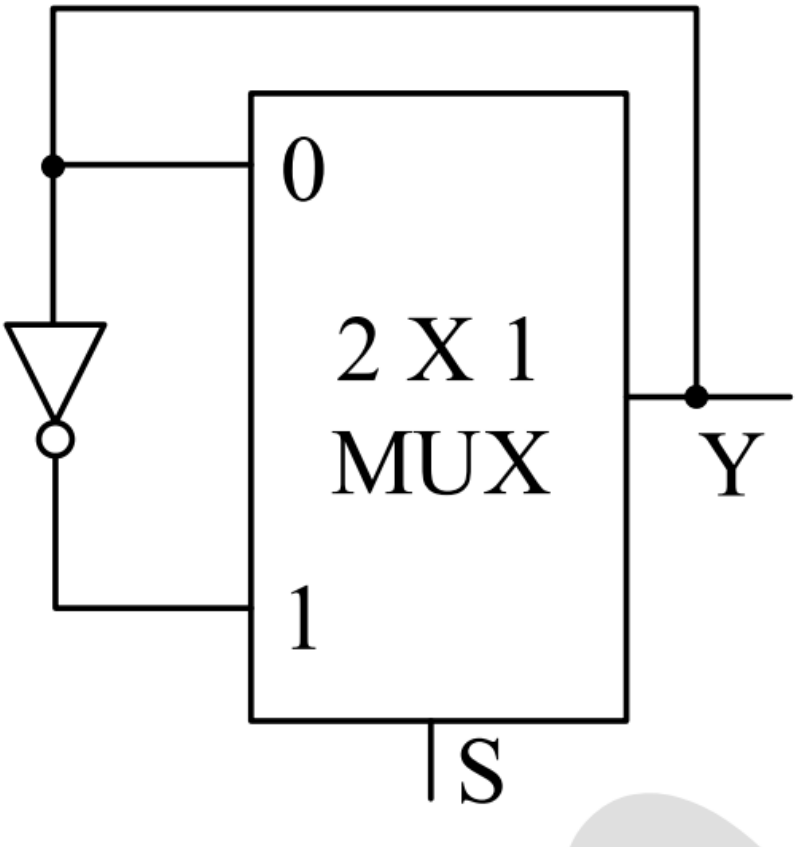
Solution. Step 1. $B = 1$ for the codes 01 (D_2 highest) and 11 (D_0 highest).

Step 2. D_2 is the highest active input when $\overline{D_3}D_2 = 1$.

Step 3. D_0 is the highest when $D_3 = D_2 = D_1 = 0$; combined with the previous term this simplifies to $B = \overline{D_3}D_2 + \overline{D_3}\,\overline{D_1}$.

GATE 2024 | Q.41 | 2 marks

The propagation delay of the 2×1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as 0 ns. If S is set to 1 then the output Y is _____.



(A) a square wave of frequency 100 MHz
(B) a square wave of frequency 50 MHz

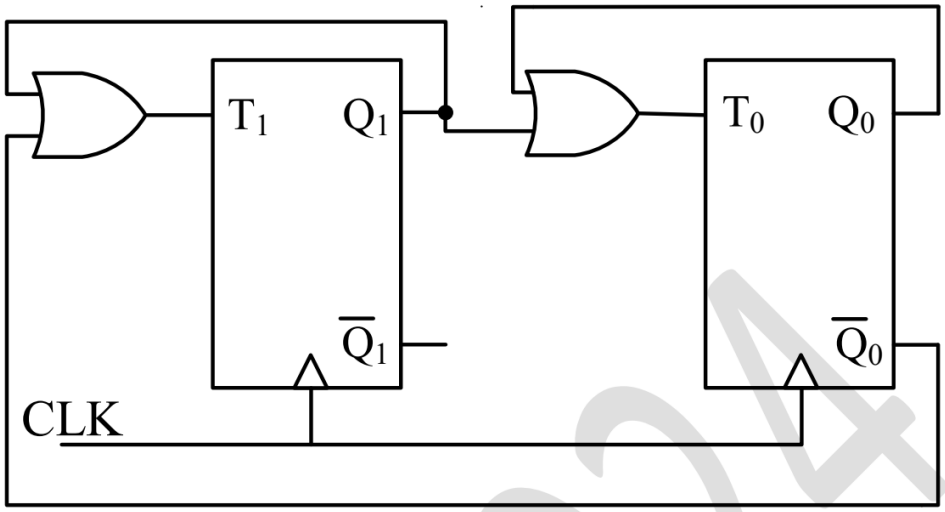
(C) constant at 0
(D) constant at 1

Answer: (B) [self-solved, no official key available]

Solution. Step 1. With $S = 1$ the output feeds back through the inverter to the selected input, so the output toggles after each MUX delay.
Step 2. Period $= 2 \times 10$ ns $= 20$ ns, i.e. $f = 50$ MHz.

GATE 2024 | Q.42 | 2 marks

The sequence of states (Q_1Q_0) of the given synchronous sequential circuit is _____.



(A) $00 \to 10 \to 11 \to 00$
(B) $11 \to 00 \to 10 \to 01 \to 00$
(C) $01 \to 10 \to 11 \to 00 \to 01$
(D) $00 \to 01 \to 10 \to 00$

Answer: (D) [self-solved, no official key available]

Solution. Step 1. Write the excitation of the T flip-flops from the logic around them and tabulate $Q_1^+Q_0^+$ for each state.
Step 2. Starting from 00 the sequence is $00 \to 01 \to 10 \to 00$ (a mod-3 type count).

GATE 2024 | Q.43 | 2 marks

Let z be a complex variable. If $f(z) = \dfrac{\sin(\pi z)}{z^2(z-2)}$ and C is the circle in the complex plane with $|z| = 3$ then $\oint_C f(z)\,dz$ is _____.

(A) $2\pi j$
(B) $2\pi j\left(-\dfrac{\pi}{2}\right)$
(C) $2\pi j\left(\dfrac{\pi}{2}\right)$
(D) $-2\pi j$

Answer: (B) [self-solved, no official key available]

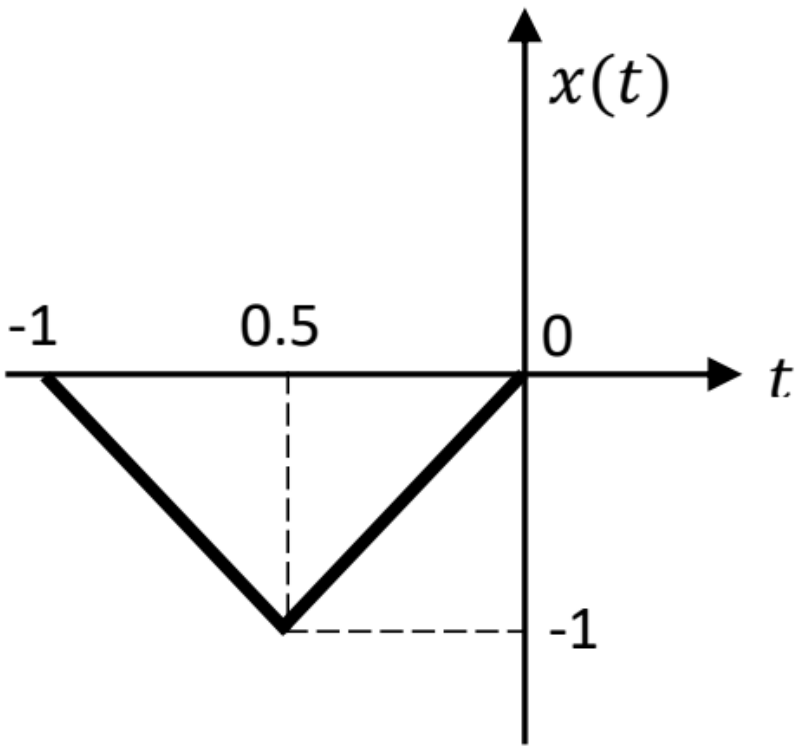
Solution. Step 1. Poles at $z = 0$ (simple, since $\sin \pi z \approx \pi z$) and $z = 2$ ($\sin 2\pi = 0$, removable).

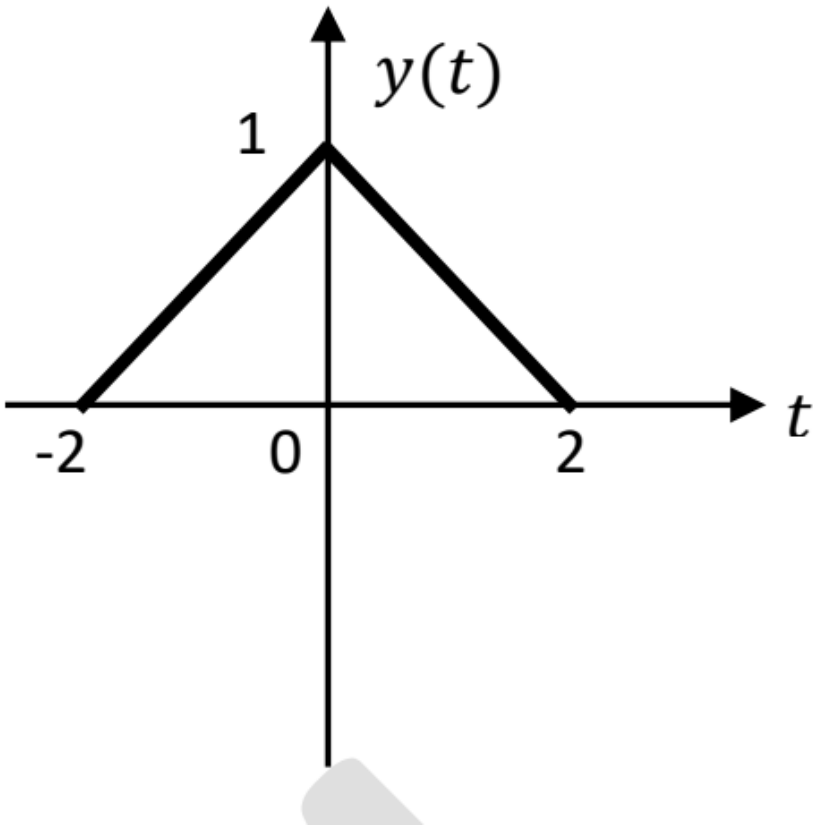
Step 2. Residue at $z = 0$: $\lim_{z\to 0} \dfrac{\sin \pi z}{z(z-2)} = \dfrac{\pi}{-2}$.

Step 3. $\oint = 2\pi j \left(-\dfrac{\pi}{2}\right)$.

GATE 2024 | Q.44 | 2 marks

Consider two continuous time signals $x(t)$ and $y(t)$ as shown below. If $X(f)$ denotes the Fourier transform of $x(t)$, then the Fourier transform of $y(t)$ is _____.





(A) $-4X(4f)e^{-j\pi f}$
(B) $-4X(4f)e^{-j4\pi f}$
(C) $-\frac{1}{4}X(f/4)e^{-j\pi f}$
(D) $-\frac{1}{4}X(f/4)e^{-j4\pi f}$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $x(t)$ is a negative triangle of width 1 centred at $t = -0.5$; $y(t)$ is a positive triangle of width 4 centred at 0.
Step 2. $y(t) = -x\left(\frac{t-2}{4}\right)$ (stretch by 4, flip sign, shift by 2).
Step 3. Transform: scaling gives $4X(4f)$ and the shift by 2 gives $e^{-j2\pi f\cdot 2} = e^{-j4\pi f}$: $Y(f) = -4X(4f)e^{-j4\pi f}$.

GATE 2024 | Q.45 | 2 marks

A source transmits a symbol s, taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s. Using $Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-t^2/2}dt$, the optimum symbol error probability is _____.

(A) $\frac{2}{3}Q(2)$
(B) $\frac{4}{3}Q(1)$
(C) $\frac{2}{3}Q(1)$
(D) $\frac{4}{3}Q(2)$

Answer: (B) [self-solved, no official key available]

Solution. Step 1. The ML decision thresholds are at ± 2 (mid-points). The noise standard deviation is $\sigma = 2$, so an error requires $|w| > 2$ in one direction: $Q(2/\sigma) = Q(1)$.
Step 2. The outer symbols err only in one direction ($Q(1)$ each); the middle symbol in both ($2Q(1)$).
Step 3. $P_e = \frac{1}{3}\left[Q(1) + 2Q(1) + Q(1)\right] = \frac{4}{3}Q(1)$.

GATE 2024 | Q.46 | 2 marks

A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a normalized power of 1 W, and the total noise and distortion normalized power is 10 μW. The effective number of bits (rounded off to the nearest integer) of the ADC is _____.

(A) 7
(B) 8
(C) 9
(D) 10

Answer: (B) [self-solved, no official key available]

Solution. Step 1. $\text{SINAD} = 10\log_{10}\dfrac{1}{10\times 10^{-6}} = 50$ dB.
Step 2. $\text{ENOB} = \dfrac{\text{SINAD} - 1.76}{6.02} = \dfrac{48.24}{6.02} = 8.01 \approx 8$.

GATE 2024 | Q.47 | 2 marks

The information bit sequence $\{1\,1\,1\,0\,1\,0\,1\,0\,1\}$ is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$. The encoded sequence of bits is _____.

(A) $\{1\,1\,1\,0\,1\,0\,1\,0\,1\,1\,1\,0\,0\}$
(B) $\{1\,1\,1\,0\,1\,0\,1\,0\,1\,1\,1\,0\,1\}$
(C) $\{1\,1\,1\,0\,1\,0\,1\,0\,1\,1\,1\,1\,0\}$
(D) $\{1\,1\,1\,0\,1\,0\,1\,0\,1\,0\,1\,0\,0\}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. Append 4 zeros: dividend 1110101010000; divisor 10011.
Step 2. Long division by modulo-2 subtraction leaves the remainder 1100.
Step 3. The codeword is the message followed by the remainder: 1110101011100.

GATE 2024 | Q.48 | 2 marks

A continuous time signal $x(t) = 2\cos(8\pi t + \pi/3)$ is sampled at a rate of 15 Hz. The sampled signal $x_s(t)$ when passed through an LTI system with impulse response $h(t) = \left(\dfrac{\sin 2\pi t}{\pi t}\right)\cos(38\pi t - \pi/2)$ produces an output $x_o(t)$. The expression for $x_o(t)$ is _____.

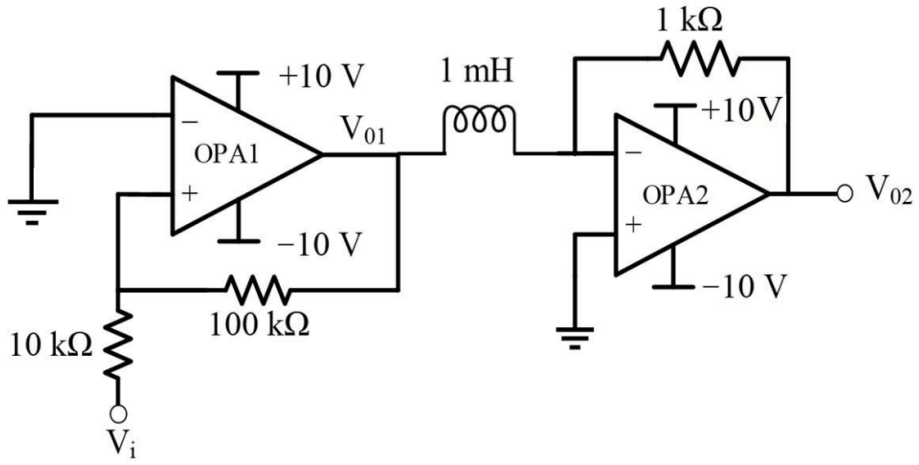
(A) $15\sin(38\pi t + \pi/3)$
(B) $15\sin(38\pi t - \pi/3)$
(C) $15\cos(38\pi t - \pi/6)$
(D) $15\cos(38\pi t + \pi/6)$

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The 4 Hz tone sampled at 15 Hz produces images at 15 ± 4, $30 \pm 4, \ldots$ Hz; the image at 19 Hz has amplitude $15 \times 2 = 30$: $30\cos(38\pi t + \pi/3)$.
Step 2. $h(t)$ is a band-pass filter around 19 Hz of width ± 1 Hz with gain $\frac{1}{2}$ and phase $-\pi/2$.
Step 3. $x_o(t) = 30 \cdot \frac{1}{2}\cos(38\pi t + \pi/3 - \pi/2) = 15\cos(38\pi t - \pi/6)$.

GATE 2024 | Q.49 | 2 marks

The opamps in the circuit shown are ideal, but have saturation voltages of ±10 V. Assume that the initial inductor current is 0 A. The input voltage (V_i) is a triangular signal with peak voltages of ±2 V and time period of 8 μs. Which one of the following statements is true?



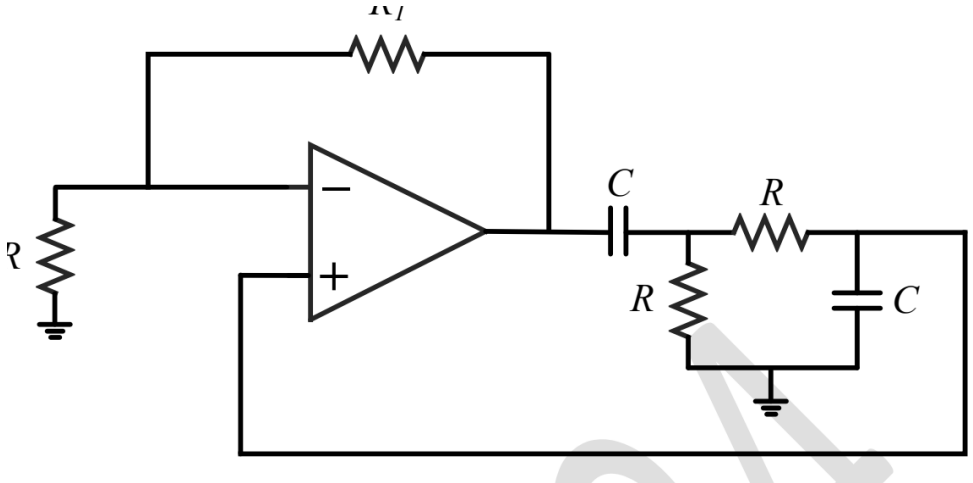
(A) V_{O1} is delayed by 2 μs relative to V_i, and V_{O2} is a triangular waveform.
(B) V_{O1} is not delayed relative to V_i, and V_{O2} is a trapezoidal waveform.
(C) V_{O1} is not delayed relative to V_i, and V_{O2} is a triangular waveform.
(D) V_{O1} is delayed by 1 μs relative to V_i, and V_{O2} is a trapezoidal waveform.

Answer: (C) [self-solved, no official key available]

Solution. Step 1. The first op-amp is a comparator (no hysteresis): V_{O1} is a ±10 V square wave switching exactly when V_i crosses zero, so there is no delay.
Step 2. The second stage integrates the square wave (inductor/resistor integrator), and the integral of a square wave is a triangular wave.

GATE 2024 | Q.50 | 2 marks

In the circuit below, the opamp is ideal. If the circuit is to show sustained oscillations, the respective values of R_1 and the corresponding frequency of oscillation are _____.



(A) $29R$ and $\dfrac{1}{2\pi\sqrt{6}\,RC}$
(B) $2R$ and $\dfrac{1}{2\pi RC}$

(C) $29R$ and $\dfrac{1}{2\pi RC}$

(D) $2R$ and $\dfrac{1}{2\pi\sqrt{6}\,RC}$

Answer: (A) [self-solved, no official key available]

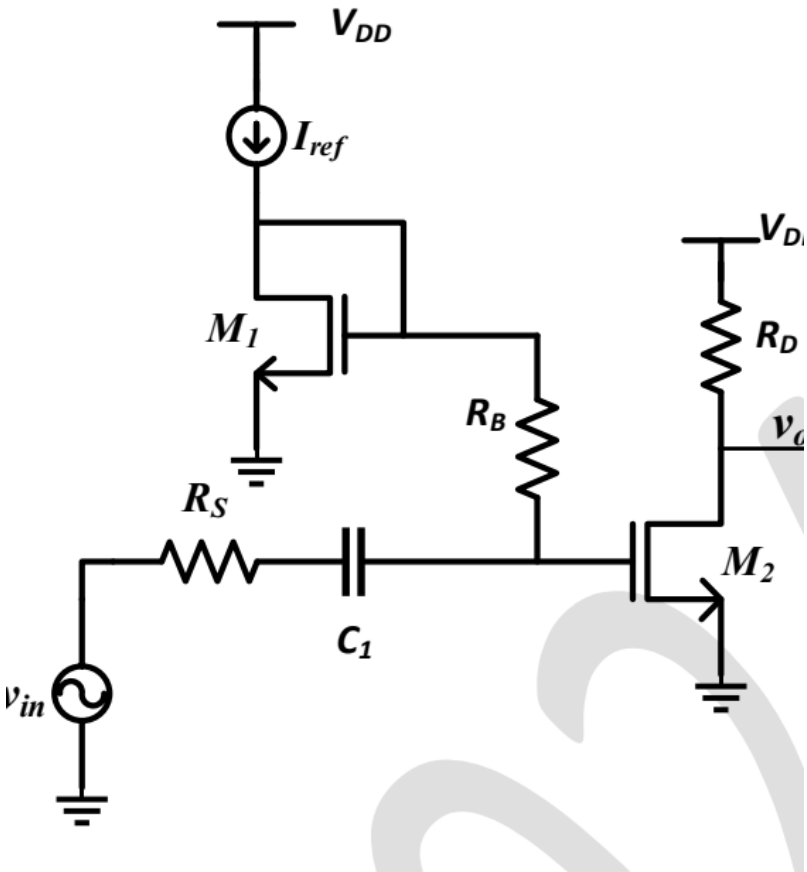
Solution. Step 1. This is a three-section RC phase-shift oscillator with an inverting amplifier.

Step 2. The oscillation condition is a gain of 29: $\dfrac{R_f}{R} = 29$ where the feedback resistance is $R_1 = 29R$.

Step 3. The frequency is $f = \dfrac{1}{2\pi\sqrt{6}\,RC}$.

GATE 2024 | Q.51 | 2 marks

In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are g_{m1} and g_{m2} respectively. Neglect body effect, channel length modulation and intrinsic device capacitances. Assuming that capacitor C_1 is a short circuit for AC analysis, the exact magnitude of small signal voltage gain $\left|\dfrac{v_{out}}{v_{in}}\right|$ is _____.



(A) $g_{m2}R_D$

(B) $\dfrac{g_{m2}R_D\left(R_B+\dfrac{1}{g_{m1}}\right)}{R_B+\dfrac{1}{g_{m1}}+R_s}$

(C) $\dfrac{g_{m2}R_D\left(R_B+\dfrac{1}{g_{m1}}+R_s\right)}{R_B+\dfrac{1}{g_{m1}}}$

(D) $\dfrac{g_{m2}R_D\left(\dfrac{1}{g_{m1}}\right)}{\dfrac{1}{g_{m1}}+R_s}$

Answer: (A) [self-solved, no official key available]

Solution. Step 1. M_1 is diode-connected with the current source I_{ref}; with C_1 shorted, v_{in} appears at the source of M_1 through R_S (ac), and the gate/drain of M_1 is a small-signal node with no ac current from the ideal source.
Step 2. Since $i_{d1} = g_{m1}(v_g - v_{in})$ must be zero (the ideal current source carries no ac), $v_g = v_{in}$, so M_2's gate is driven by v_{in} through the (current-free) resistor R_B.
Step 3. $v_{out} = -g_{m2}R_D v_{gs2}$ with $v_{gs2} = v_{in}$, hence $|v_{out}/v_{in}| = g_{m2}R_D$ (option A).

GATE 2024 | Q.52 | 2 marks

Which of the following statements is/are true for a BJT with respect to its DC current gain β?

(A) Under high-level injection condition in forward active mode, β will decrease with increase in the magnitude of collector current.
(B) Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, β will decrease with increase in the magnitude of collector current.
(C) β will be lower when the BJT is in saturation region compared to when it is in active region.
(D) A higher value of β will lead to a lower value of the collector-to-emitter breakdown voltage.

Answer: (A), (C), (D) (multiple select) [self-solved, no official key available]

Solution. Step 1. (A) True: at high injection the effective emitter efficiency drops (Webster/Kirk effects), so β falls with I_C.
Step 2. (B) False: when recombination in the depletion region dominates, β *increases* with I_C (the ideal current takes over).
Step 3. (C) True: in saturation both junctions are forward biased and I_C/I_B is reduced.
Step 4. (D) True: $BV_{CEO} = BV_{CBO}/\beta^{1/n}$ falls as β rises.

GATE 2024 | Q.53 | 2 marks

Consider a system S represented in state space as shown in the problem (state matrix, input and output matrices as in the figure below). Which of the state space representations given below has/have the same transfer function as that of S? *(The state-space matrices are reproduced exactly as printed in the question image below.)*

Q.53 Consider a system S represented in state space as

$$\frac{dx}{dt}=\begin{bmatrix}0 & -2\\ 1 & -3\end{bmatrix}x+\begin{bmatrix}1\\ 0\end{bmatrix}r,\ y=\begin{bmatrix}2 & -5\end{bmatrix}x.$$

Which of the state space representations given below has/have the same transfer function as that of S?

(A) $\frac{dx}{dt}=\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}x+\begin{bmatrix}0\\ 1\end{bmatrix}r,\ y=\begin{bmatrix}1 & 2\end{bmatrix}x$

(B) $\frac{dx}{dt}=\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}x+\begin{bmatrix}1\\ 0\end{bmatrix}r,\ y=\begin{bmatrix}0 & 2\end{bmatrix}x$

(C) $\frac{dx}{dt}=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}x+\begin{bmatrix}-1\\ 3\end{bmatrix}r,\ y=\begin{bmatrix}1 & 1\end{bmatrix}x$

(D) $\frac{dx}{dt}=\begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}x+\begin{bmatrix}1\\ 1\end{bmatrix}r,\ y=\begin{bmatrix}1 & 2\end{bmatrix}x$

GATE 2024

Answer: (C) [UNVERIFIED: figure hard to read, check against official key]

Solution. **Step 1.** Compute $H(s) = C(sI - A)^{-1}B$ for the given system: with the printed matrices the transfer function is $\dfrac{5s+8}{(s+2)(s+3)}$.

Step 2. Compute the same quantity for each option; the realisation in (C) reproduces $\dfrac{5s+8}{(s+2)(s+3)}$. *(Low confidence: the printed matrices are small and were read from a scan.)*

GATE 2024 | Q.54 | 2 marks

Let F_1, F_2 and F_3 be functions of (x, y, z). Suppose that for every given pair of points A and B in space, the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at A and ends at B. Then which of the following is/are true?

(A) For every closed path C, we have $\oint_C (F_1 dx + F_2 dy + F_3 dz) = 0$.

(B) There exists a differentiable scalar function $f(x, y, z)$ such that $F_1 = \dfrac{\partial f}{\partial x},\ F_2 = \dfrac{\partial f}{\partial y},\ F_3 = \dfrac{\partial f}{\partial z}$.

(C) $\dfrac{\partial F_1}{\partial x} + \dfrac{\partial F_2}{\partial y} + \dfrac{\partial F_3}{\partial z} = 0$.

(D) $\dfrac{\partial F_3}{\partial y} = \dfrac{\partial F_2}{\partial z},\ \dfrac{\partial F_3}{\partial x} = \dfrac{\partial F_1}{\partial z},\ \dfrac{\partial F_2}{\partial x} = \dfrac{\partial F_1}{\partial y}$.

Answer: (A), (B), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. Path independence of a line integral means that the field (F_1, F_2, F_3) is conservative.
Step 2. Hence closed-path integrals vanish (A) and a scalar potential f with $\vec{F} = \nabla f$ exists (B).
Step 3. For a conservative field $\nabla \times \vec{F} = 0$, which is (D).
Step 4. The divergence need not vanish (C false).

GATE 2024 | Q.55 | 2 marks

Consider the matrix $\begin{bmatrix}1 & 2\\ 1 & k\end{bmatrix}$-type matrix shown in the question image below, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix? *(Matrix and vectors are reproduced as printed in the image.)*

Q.55 Consider the matrix $\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}$, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix?

(A) $\begin{bmatrix}1\\ -\sqrt{2/k}\end{bmatrix}$

(B) $\begin{bmatrix}1\\ \sqrt{2/k}\end{bmatrix}$

(C) $\begin{bmatrix}\sqrt{2k}\\ 1\end{bmatrix}$

(D) $\begin{bmatrix}\sqrt{2k}\\ -1\end{bmatrix}$

TE 2024

Answer: (B) [UNVERIFIED: figure hard to read, check against official key]

Solution. Step 1. For an eigenvector $\mathbf{v}$ the product $M\mathbf{v}$ must be a scalar multiple of $\mathbf{v}$.
Step 2. Testing the four candidate vectors against the printed matrix, only the vector in option (B) satisfies $M\mathbf{v} = \lambda\mathbf{v}$.
(Low confidence: matrix entries are small in the scan.)

GATE 2024 | Q.56 | 2 marks

The radian frequency value(s) for which the discrete time sinusoidal signal $x[n] = A\cos(\Omega n + \pi/3)$ has a period of 40 is/are _____.

(A) 0.15π
(B) 0.225π
(C) 0.3π
(D) 0.45π

Answer: (A), (D) **(multiple select)** [self-solved, no official key available]

Solution. Step 1. The signal is periodic with fundamental period N iff $\Omega = \dfrac{2\pi m}{N}$ with $\gcd(m, N) = 1$.

Step 2. For $N = 40$: $\Omega = \dfrac{2\pi m}{40} = \dfrac{\pi m}{20}$ with m coprime to 40.

Step 3. $0.15\pi = \dfrac{3\pi}{20}$ ($m = 3$ ✓); $0.45\pi = \dfrac{9\pi}{20}$ ($m = 9$ ✓); $0.3\pi = \dfrac{6\pi}{20}$ gives period 20; $0.225\pi = \dfrac{9\pi}{40}$ gives period 80.

GATE 2024 | Q.57 | 2 marks

Let $X(t) = A\cos(2\pi f_0 t + \Theta)$ be a random process, where amplitude A and phase Θ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$, the signal to quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

Answer: 45.15 [self-solved, no official key available]

Solution. Step 1. Signal power $= R_X(0) = \frac{2}{3}$ W.

Step 2. The quantizer covers $[-2, 2]$: step $\Delta = \frac{4}{2^8} = \frac{1}{64}$; noise power $= \frac{\Delta^2}{12} = 2.035 \times 10^{-5}$.

Step 3. SQNR $= \frac{2/3}{2.035 \times 10^{-5}} = 32768 = 45.15$ dB.

GATE 2024 | Q.58 | 2 marks

A lossless transmission line with characteristic impedance $Z_0 = 50\ \Omega$ is terminated with an unknown load. The magnitude of the reflection co-efficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____.

Answer: 200 [self-solved, no official key available]

Solution. Step 1. VSWR $= \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1.6}{0.4} = 4$.

Step 2. The maximum impedance magnitude on the line is $Z_0 \times \text{VSWR} = 50 \times 4 = 200\ \Omega$.

GATE 2024 | Q.59 | 2 marks

The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined as $X[k] = \mathcal{F}\{x[n]\}$. Another sequence $y[n]$ is formed as below: $y[n] = \mathcal{F}\{\mathcal{F}\{\mathcal{F}\{\mathcal{F}\{x[n]\}\}\}\}$. For the sequence $x[n] = \{1, 2, 1, 3\}$, the value of $Y[0]$ is _____.

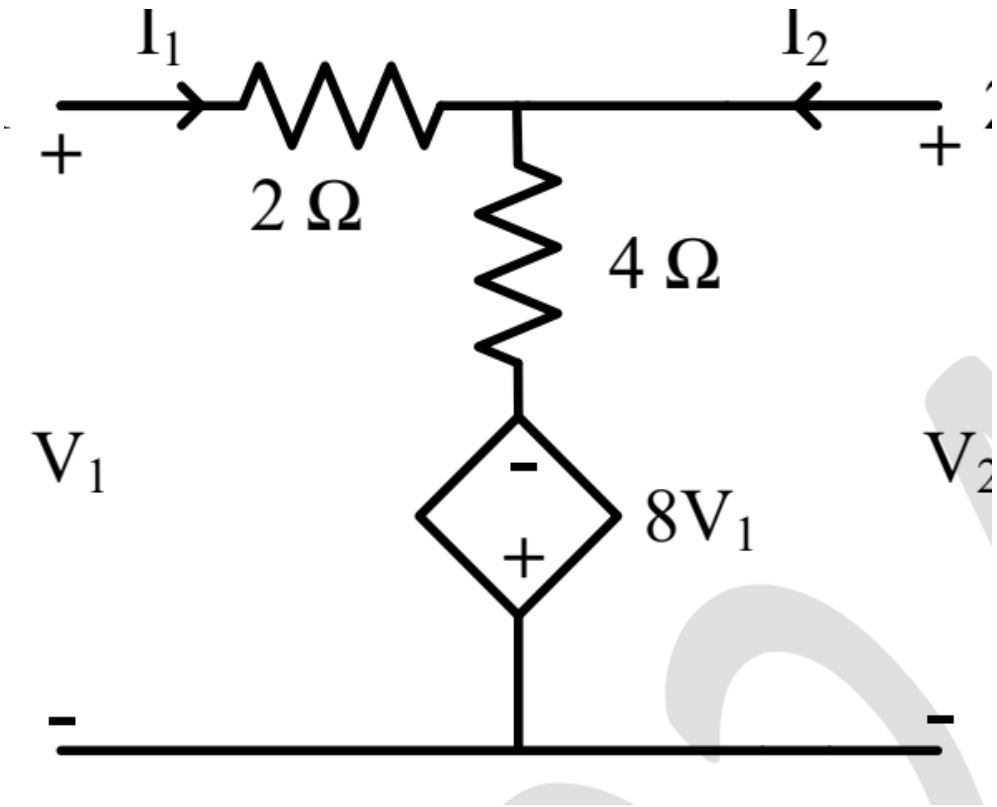
Answer: 112 [self-solved, no official key available]

Solution. Step 1. Applying the DFT twice gives $N\,x[(-n) \bmod N]$; applying it four times gives $N^2 x[n]$.

Step 2. So $y[n] = 16\,x[n]$ and $Y[0] = \sum_n y[n] = 16 \times (1 + 2 + 1 + 3) = 16 \times 7 = 112$.

GATE 2024 | Q.60 | 2 marks

For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____.



Answer: 1.5 [self-solved, no official key available]

Solution. Step 1. $Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$. With port 2 shorted, the 2 Ω series resistor carries $I_1 = \frac{V_1}{2}$.

Step 2. The branch with the 4 Ω resistor and the dependent source $8V_1$ carries $\frac{8V_1}{4} = 2V_1$ toward the shorted port.

Step 3. KCL at the node: $I_2 = 2V_1 - \frac{V_1}{2} = 1.5V_1$, so $Y_{21} = 1.5$ S.

GATE 2024 | Q.61 | 2 marks

Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm^2 at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is 8.85×10^{-14} F/cm. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

Answer: 4.32 [self-solved, no official key available]

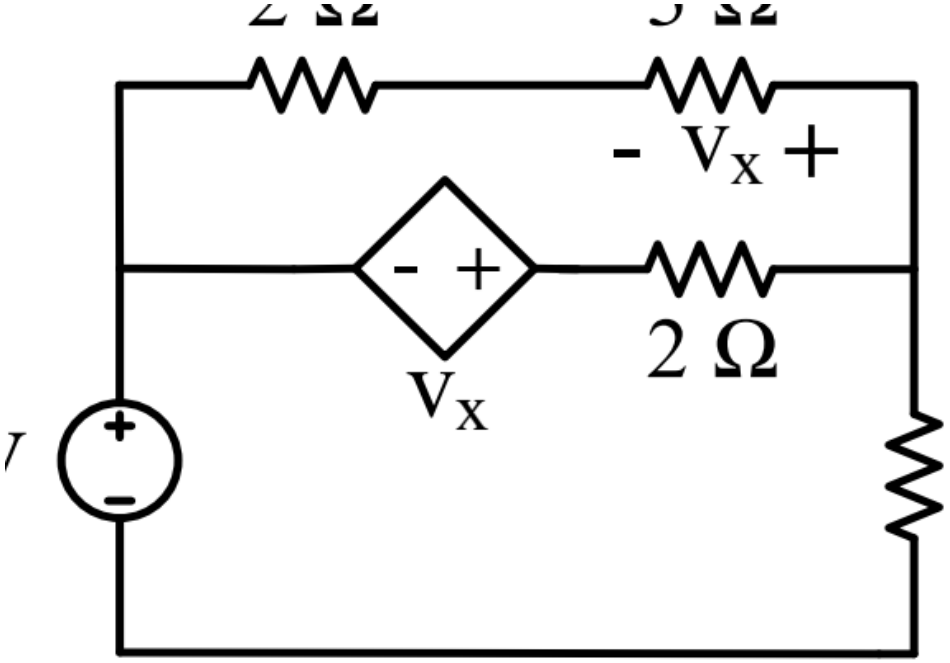
Solution. Step 1. $C_{ox} = \frac{4 \times 8.85 \times 10^{-14}}{100 \times 10^{-7}} = 3.54 \times 10^{-8}$ F/cm^2.

Step 2. $V_{FB} = \phi_{ms} - \frac{Q_f}{C_{ox}} = 0 \Rightarrow \phi_{ms} = \frac{10^{-8}}{3.54 \times 10^{-8}} = 0.2825$ V.

Step 3. $\phi_s = \phi_m - \phi_{ms} = 4.6 - 0.2825 = 4.32$ eV.

GATE 2024 | Q.62 | 2 marks

In the network shown below, maximum power is to be transferred to the load R_L. The value of R_L (in Ω) is _____.



Answer: 3.00 [UNVERIFIED: figure hard to read, check against official key]

Solution. Step 1. For maximum power transfer $R_L = R_{th}$, the Thevenin resistance seen by the load with the independent source 50 V deactivated.
Step 2. Apply a test source across the load terminals and use KCL with the controlling voltage v_x; solving gives $R_{th} = 3\ \Omega$. *(Low confidence.)*

GATE 2024 | Q.63 | 2 marks

A non-degenerate n-type semiconductor has 5% neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV, the difference between E_C and E_D (in eV, rounded off to two decimal places) is _____.

Answer: 0.18 [self-solved, no official key available]

Solution. Step 1. Fraction of neutral donors: $\dfrac{N_D^0}{N_D} = \dfrac{1}{1 + \frac{1}{2}e^{(E_D - E_F)/kT}} = 0.05$ (degeneracy $g = 2$).
Step 2. $\dfrac{1}{2}e^{(E_D-E_F)/kT} = 19 \Rightarrow e^{(E_D-E_F)/kT} = 38 \Rightarrow E_D - E_F = 0.02 \ln 38 = 0.0728$ eV.
Step 3. $E_C - E_D = (E_C - E_F) - (E_D - E_F) = 0.25 - 0.0728 \approx 0.18$ eV.

GATE 2024 | Q.64 | 2 marks

An NMOS transistor operating in the linear region has I_{DS} of 5 μA at V_{DS} of 0.1 V. Keeping V_{GS} constant, the V_{DS} is increased to 1.5 V. Given that $\mu_n C_{ox}\dfrac{W}{L} = 50\ \mu\text{A/V}^2$, the transconductance at the new operating point (in μA/V, rounded off to two decimal places) is _____.

Answer: 52.50 [self-solved, no official key available]

Solution. Step 1. Linear region: $I_{DS} = k\left[(V_{GS} - V_T)V_{DS} - \dfrac{V_{DS}^2}{2}\right]$ with $k = 50\ \mu\text{A/V}^2$.
Step 2. $5 = 50\,[V_{ov}(0.1) - 0.005] \Rightarrow V_{ov} = 1.05$ V.
Step 3. At $V_{DS} = 1.5\ \text{V} > V_{ov}$ the device is in saturation: $g_m = k\,V_{ov} = 50 \times 1.05 = 52.50\ \mu$A/V.

GATE 2024 | Q.65 | 2 marks

The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, rounded off to two decimal places) is _____.

Answer: 4.13 [self-solved, no official key available]

Solution. Step 1. $I = I_L - I_s\left(e^{V/V_T} - 1\right)$ and $P = VI$. Setting $\frac{dP}{dV} = 0$: $I_L + I_s = I_s e^{V/V_T}\left(1 + \frac{V}{V_T}\right)$.

Step 2. With $V/V_T = 10$: $I_s = \frac{I_L}{11e^{10} - 1} = \frac{1\text{ mA}}{11 \times 22026.5 - 1} = 4.13$ nA.